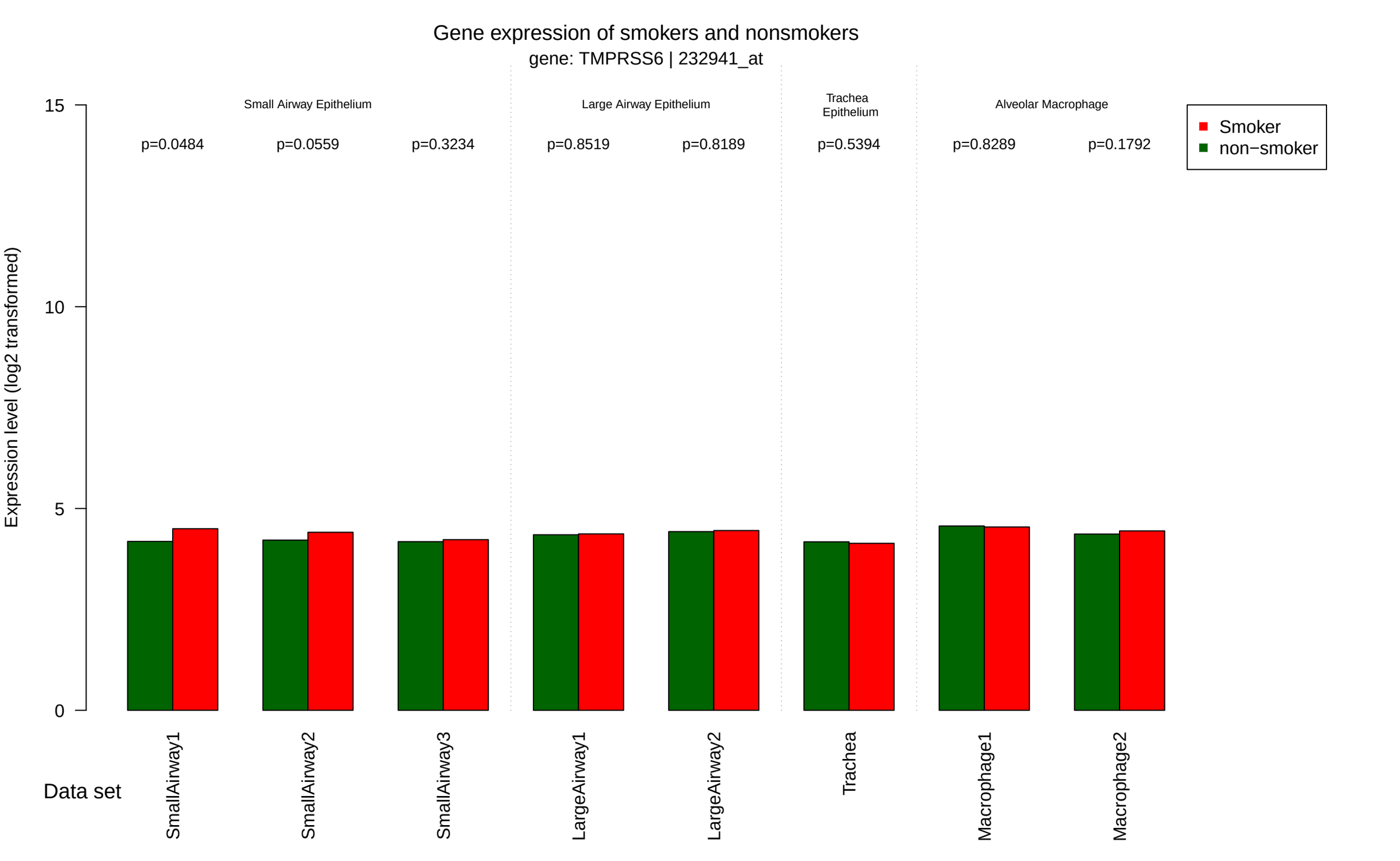
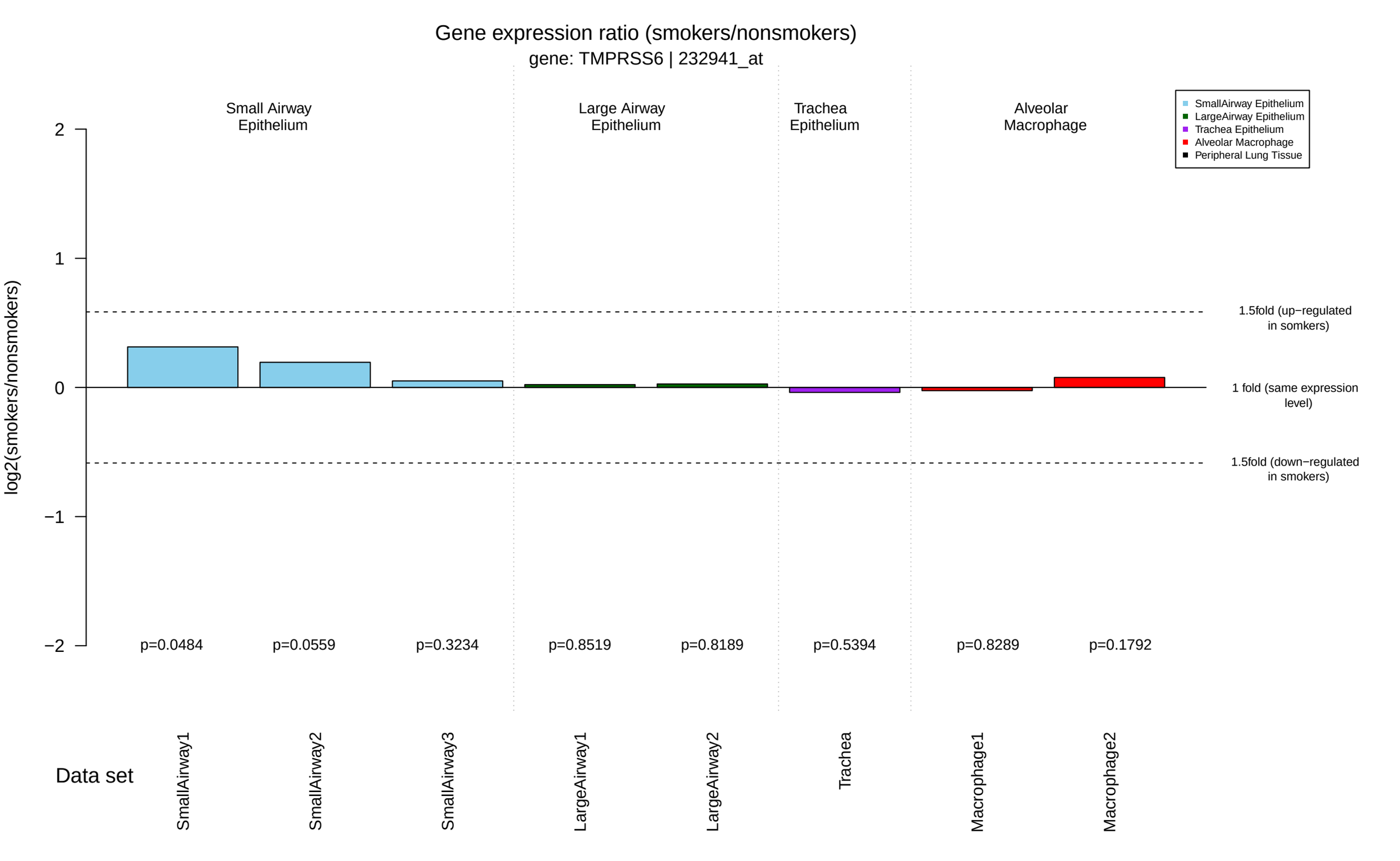
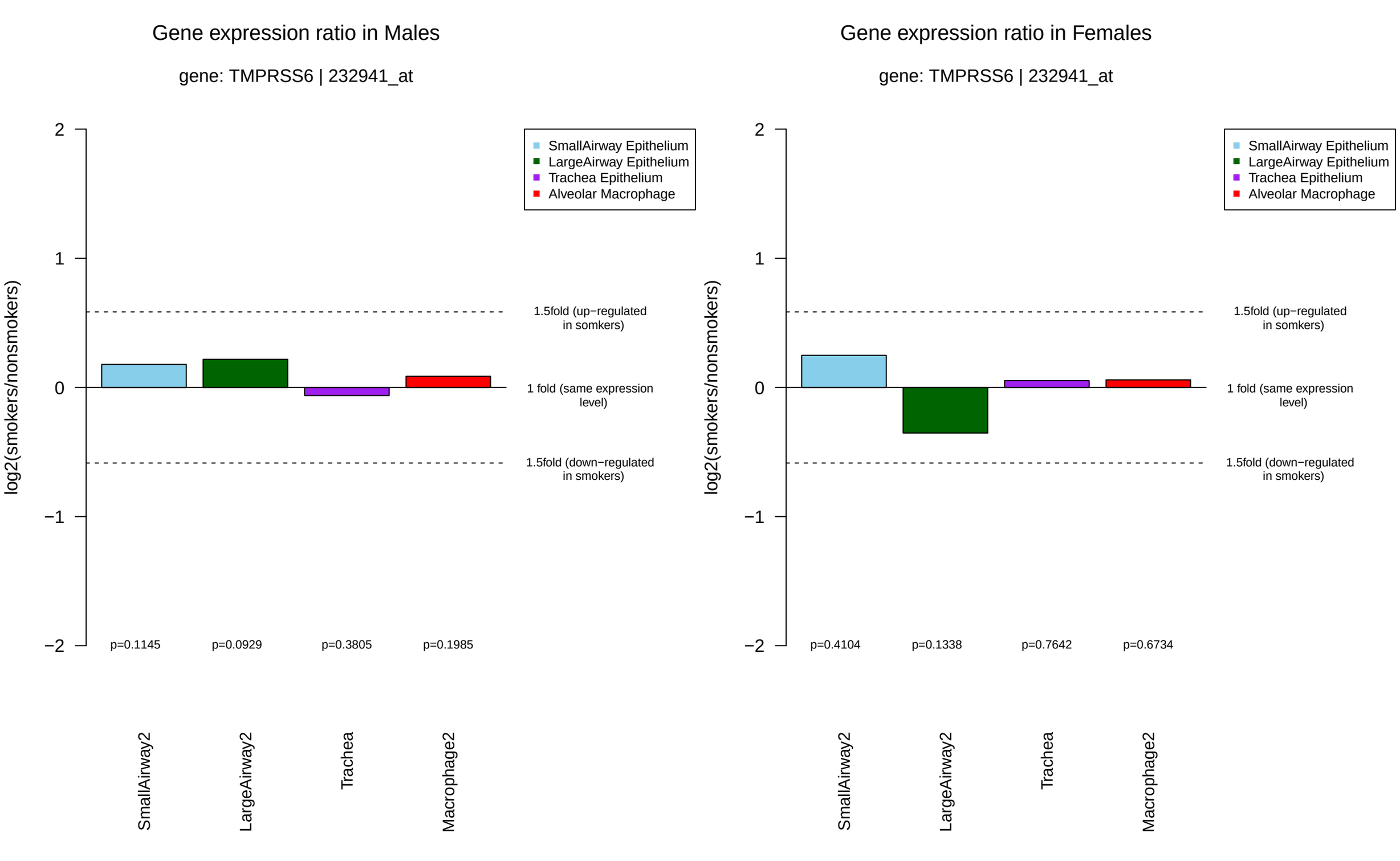
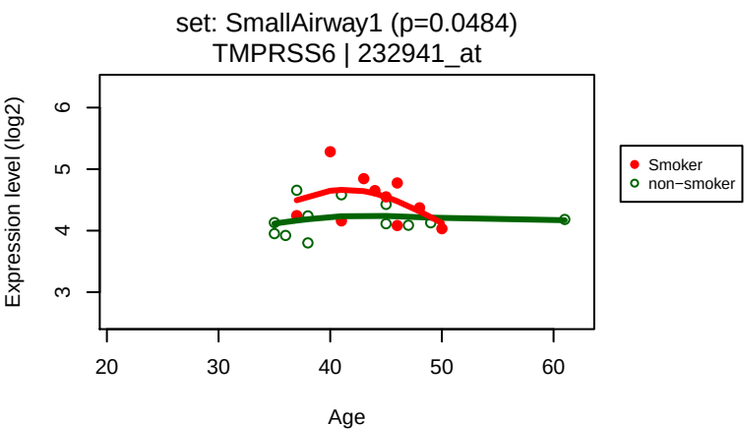
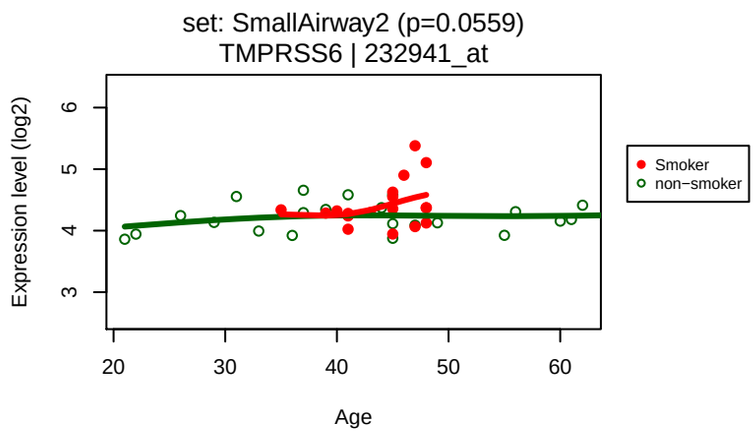
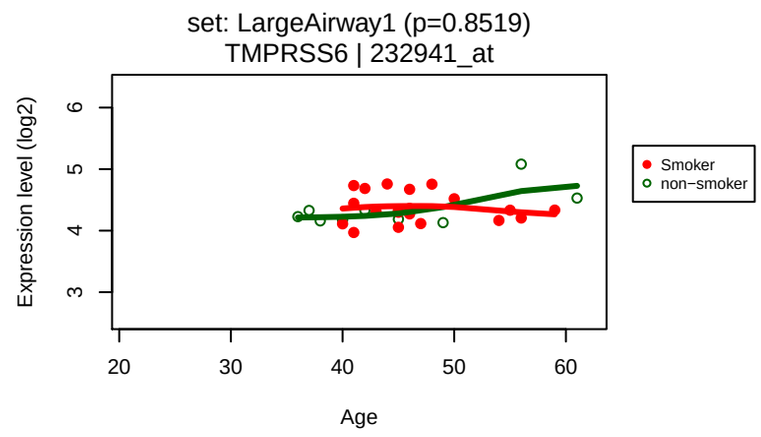

Gene expression of smokers and nonsmokers
gene: TMPRSS6 | 232941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0484
p=0.0559
p=0.3234
p=0.8519
p=0.8189
p=0.5394
p=0.8289
p=0.1792
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

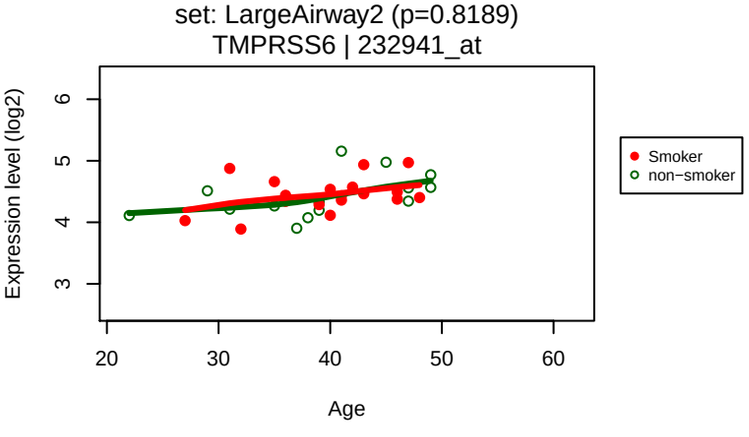

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS6 | 232941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0484
p=0.0559
p=0.3234
p=0.8519
p=0.8189
p=0.5394
p=0.8289
p=0.1792
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

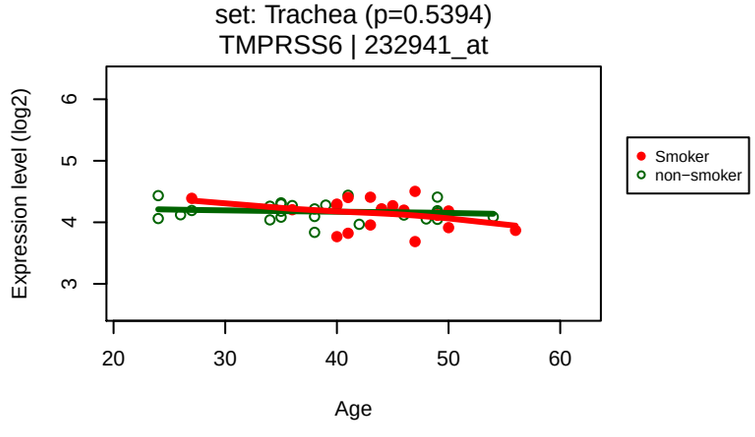

Gene expression ratio in Males
gene: TMPRSS6 | 232941_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1145
p=0.0929
p=0.3805
p=0.1985
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS6 | 232941_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4104
p=0.1338
p=0.7642
p=0.6734
SmallAirway2
LargeAirway2
Trachea
Macrophage2

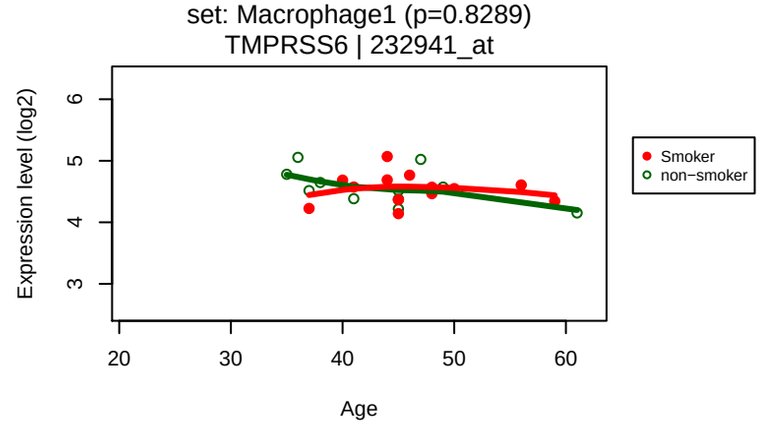

set: SmallAirway1 (p=0.0484)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker

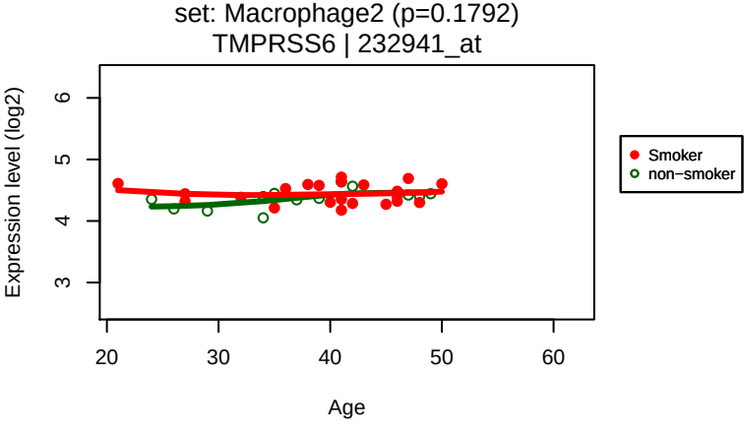

set: SmallAirway2 (p=0.0559)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker

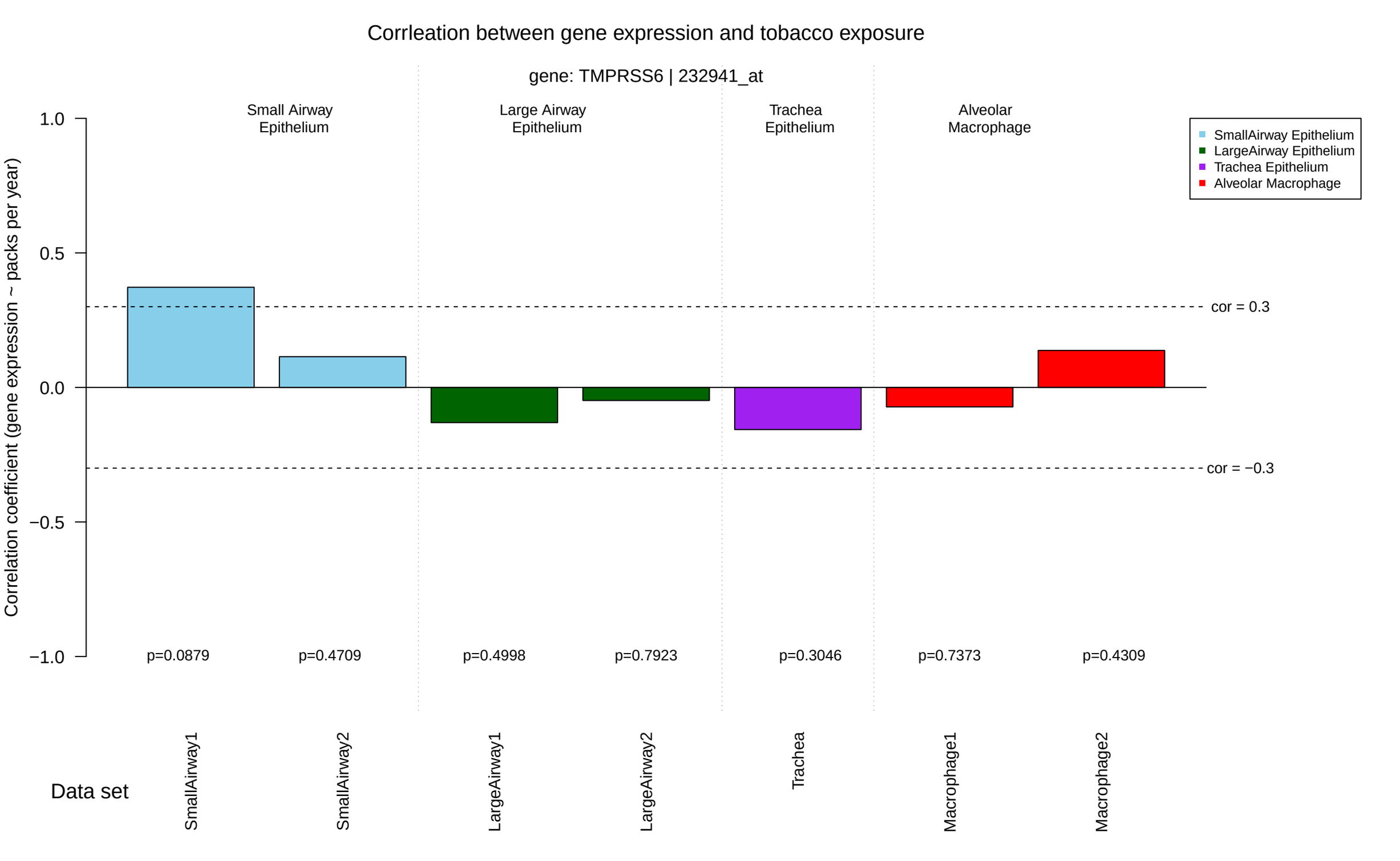

set: LargeAirway1 (p=0.8519)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8189)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5394)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8289)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1792)
TMPRSS6 | 232941_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS6 | 232941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0879
p=0.4709
p=0.4998
p=0.7923
p=0.3046
p=0.7373
p=0.4309
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2